1 (a) Expand $7(2x + 7)$

$14x + 49$

(1)

(b) Factorise $3y + 12$

$3(y + 4)$

(1)

(Total for Question 1 is 2 marks)

---

2 (a) Expand $5a(a - 6)$

$5a^2 - 30a$

(2)

(b) Solve $4(b + 2) = 24$

$4b + 8 = 24$

$4b = 16$

$b = 4$

$b =$ 4

(2)

(Total for Question 2 is 4 marks)

---

3 (a) Factorise fully $12m + 8m^2$

$4m(3 + 2m)$

(2)

(b) Solve $3(n - 5) = 27$

$3n - 15 = 27$

$3n = 42$

$n = 14$

$n =$ 14

(2)

(Total for Question 3 is 4 marks)

4 (a) Expand $8(3s - 2)$

$24s - 16$

(1)

(b) Factorise $4t + 20$

$4(t + 5)$

(1)

(Total for Question 4 is 2 marks)

---

5 (a) Factorise fully $5a^2b + 15ab^2$

$5ab(a + 3b)$

(2)

(b) Solve $6(c - 8) = 42$

$6c - 48 = 42$

$6c = 90$

$c = 15$

$c =$ 15

(2)

(Total for Question 5 is 4 marks)

---

6 (a) Factorise $18x + 24$

$6(3x + 4)$

(1)

(b) Expand $3(2y - 4)$

$6y - 12$

(1)

(Total for Question 6 is 2 marks)

7 (a) Expand $p(p-3)$

$p^2 - 3p$

(1)

(b) Factorise $16q + 8$

$8(2q+1)$

(1)

(Total for Question 7 is 2 marks)

---

8 (a) Factorise fully $6x^2 - 4xy$

$2x(3x-2y)$

(2)

(b) Solve $2(w-4) = 13$

$2w - 8 = 13$

$2w = 21$

$w = \frac{21}{2}$

$w =$ $\frac{21}{2}$ or 10.5

(2)

(Total for Question 8 is 4 marks)

---

9 (a) Factorise $x^2 - 9x$

$x(x-9)$

(1)

(b) Expand $6(5y+1)$

$30y + 6$

(1)

(Total for Question 9 is 2 marks)

10 (a) Expand $3(5x - 8)$

$15x - 24$

(1)

(b) Factorise $18y + 15$

$3(6y + 5)$

(1)

(Total for Question 10 is 2 marks)

---

11 (a) Expand $7(2h - 3)$

$14h - 21$

(1)

(b) Expand and Simplify $4(g + 5) + 3(g - 2)$

$4g + 20 + 3g - 6$

$7g + 14$

(2)

(Total for Question 11 is 3 marks)

---

12 (a) Factorise fully $7xy + 21x$

$7x(y + 3)$

(2)

(b) Solve $6(p + 3) = 42$

$6p + 18 = 42$

$6p = 24$

$p = 4$

$p =$ 4

(2)

(Total for Question 12 is 4 marks)

13 (a) Expand $a(a + b)$

$a^2 + ab$

(1)

(b) Factorise $15y - 6$

$3(5y - 2)$

(1)

(Total for Question 13 is 2 marks)

---

14 (a) Expand $9x(3y - 8)$

$27xy - 72x$

(2)

(b) Expand and Simplify $7(t - 4) + 5(t - 2)$

~~$7t - 28 + 5t - 10$~~

$7t - 28 + 5t - 10$

$12t - 38$

(2)

(Total for Question 14 is 4 marks)

---

15 (a) Factorise fully $30x^3 + 12x$

$6x(5x^2 + 2)$

(2)

(b) Solve $5(f - 2) = 22$

$5f - 10 = 22$

$5f = 32$

$f = \frac{32}{5}$

$f = \frac{32}{5}$ OR 6.4

(2)

(Total for Question 15 is 4 marks)

16 (a) Expand $x(8x + 1)$

$8x^2 + x$

(1)

(b) Factorise $18 + 63y$

$9(2 + 7y)$

(1)

(Total for Question 16 is 2 marks)

---

17 (a) Expand $2x^2(4x - 9)$

$8x^3 - 18x^2$

(2)

(b) Expand and Simplify $6(y + 3) - 5(y - 4)$

$6y + 18 - 5y + 20$

$y + 38$

(2)

(Total for Question 17 is 4 marks)

---

18 (a) Factorise fully $30a^2 + 40ab$

$10a(3a + 4b)$

(2)

(b) Solve $3(g + 9) = 21$

$3g + 27 = 21$

$3g = -6$

$g = -2$

$g =$ -2

(2)

(Total for Question 18 is 4 marks)

19 (a) Expand $n(5n + 1)$

$5n^2 + n$

(1)

(b) Factorise $18m + mn$

$m(18 + n)$

(1)

(Total for Question 19 is 2 marks)

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20 (a) Expand $3x(7x^2 - y)$

$21x^3 - 3xy$

(2)

(b) Expand and Simplify $3(6y + 5) - 2(4y - 1)$

$18y + 15 - 8y + 2$

$10y + 17$

(2)

(Total for Question 20 is 4 marks)

---

21 (a) Factorise fully $18a^2bc + 30abc^2$

$6abc(3a + 5c)$

(2)

(b) Expand and Simplify $4(2y - 7) - 3(5y - 3)$

$8y - 28 - 15y + 9$

$-7y - 19$

(2)

(Total for Question 21 is 4 marks)